THE COUNIT OF A QUASI-COMPACT SEPARATED ETALE MORPHISM OF ALGEBRAIC SPACES IS A SPLIT SURJECTION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-spaces-cohomology/preamble.tex is excerpted verbatim as three spans, lines 8-12 (verbatim package and the project's own reference/slogan/history display environments), lines 14-18 (Xy-pic setup) and lines 154-217 (theorem declarations and mathematical macros). The wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler. No macro was invented or redefined.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- The projection formula identifying the pullback of a pushforward along the fibre square. Same chapter spaces-cohomology.tex, label lemma-affine-base-change (invoked at proof line 2036). Applied to the cartesian square with the two projections p and q, it gives $f^*f_*\mathcal{F} = q_*p^*\mathcal{F}$, which is the sheaf whose relation to $\mathcal{F}$ is being described.
- The diagonal of a separated etale morphism is an open and closed immersion. Standard: $\Delta : X \to X \times_Y X$ is an open immersion because f is etale and a closed immersion because f is separated (used at proof lines 2038-2039). It splits the structure sheaf of the fibre square into the diagonal and the complementary open and closed piece, which is what makes $p^*\mathcal{F}$ decompose.

Source reference. Original lemma 07U8: sources/stacks-spaces-cohomology/pinned-original-source.tex, source lines 1902–1902. Navigation pointer for \ref{lemma-affine-base-change} used inside the counted span (line 2036); the argument that consumes it is retained in the span.

Lemma 0.1. *Let $f : X \to Y$ be a quasi-compact, separated, étale morphism of algebraic spaces. Then for any quasi-coherent $\mathcal{O}_X$-module $\mathcal{F}$ the map $f^*f_*\mathcal{F} \to \mathcal{F}$ is split.*

Date: stacks-project:spaces-cohomology.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. Consider the cartesian diagram

$$\begin{array}{ccc} X \times_Y X & \xrightarrow{p} & X \\ {\scriptstyle q}\downarrow & & \downarrow{\scriptstyle f} \\ X & \xrightarrow{f} & Y \end{array}$$

By Lemma 07U8 we have $f^*f_*\mathcal{F} = q_*p^*\mathcal{F}$. The morphism $\Delta : X \to X \times_Y X$ is open (as a morphism between algebraic spaces étale over Y) and closed as f is separated. Thus we see that $p^*\mathcal{F}$ is a direct sum of $\Delta_*\mathcal{F}$ and a quasi-coherent module supported on the (open and closed) complement of $\Delta(X)$. Tracing the maps the reader verifies that

$$\mathcal{F} = q_*\Delta_*\mathcal{F} \to q_*p^*\mathcal{F} = f^*f_*\mathcal{F} \to \mathcal{F}$$

is the identity map. Details omitted. □